

Process Control: Level Control

Final Lab Report
Unit Operations Lab 1
January 27, 2026

Group 3, Tuesday 9 AM

Noah Guale

Jazmin Aca

Angelica Ramirez

Basil Hashim

Hiba-Batool Naqvi

Table of Contents

1. Executive Summary.....	3
2. Introduction.....	4
3. Theory.....	5
4. Results.....	7
5. Discussion.....	15
6. References.....	17
7. Appendix I: Data Tabulation/Graphs.....	19
8. Appendix II: Error Analysis.....	19
9. Appendix III: Sample Calculations.....	20
10. Appendix IV: Individual Team Contributions.....	20

1. Executive Summary

This experiment evaluated manual, proportional (P), and proportional Integral (PI) control strategies for regulating liquid level in a tank. The objective was to determine how each control method affects steady state offset, stability, and disturbance response during filling, draining, and steady water level operation.

Manual control was first implemented by adjusting pump speed to achieve and maintain a steady water level while applying inflow and outflow disturbances using solenoid valve. Manual operation required frequent adjustments and was sensitive to disturbances, demonstrating the limitations of open-loop control.

Proportional control was then applied at multiple proportional band settings (200%, 100%, 50%, 20%) with setpoints of 75 mm and 100 mm. Higher proportional bands produced stable behavior but large steady-state offsets, while lower bands reduced offsets but introduced oscillations. A proportional band near 50% provided the best balance between stability and offset.

Finally, integral control (PI) was added at a fixed proportional band (20%) to reduce offset. Results showed reduced steady-state error, however, lower integral times caused increased oscillations and occasional instability when the system was under disturbance and higher setpoints. The results show that while PI control improves accuracy by reducing offset, both P and I values need to be carefully selected to achieve a stable and reliable level control.

2. Introduction

The objective of this experiment is to evaluate different liquid-level control strategies (manual, P, and PI) to maintain a steady water level in a tank system. Liquid-level control is a fundamental problem in process control engineering as it is commonly found in chemical, biochemical, and industrial fluid systems such as storage tanks, separation units, and reactors. Maintaining a stable liquid level is important for ensuring safe operation, consistent process performance, and product quality [1]. Variations in inflow, outflow, or system disturbances can significantly affect level stability, making effective control strategies essential.

Proportional (P) control is widely used due to its simplicity and ease of implementation. In P control, the controller output is directly proportional to the error between the measured process variable and the desired setpoint. While P control can provide rapid response, it typically results in a steady state offset, particularly in systems with continuous disturbances. To reduce this offset, integral action is introduced to improve steady state accuracy. However, improper tuning of PI controllers can introduce oscillations and instability [2].

The apparatus used in this experiment is the Armfield PCT50 Level Control Process Unit, in which water was pumped from a sump tank into a process vessel using a variable speed centrifugal pump. Water exits the vessel through orifices and a solenoid valve that can be opened or closed to introduce controlled disturbances. The liquid level in the vessel is measured using a pressure sensor and all measurements and controller actions are recorded digitally. Experiments were performed by manually adjusting the pump speed, implementing proportional control with varying proportional band values,

and applying proportional integral control with different integral time settings to evaluate system stability, offset, and disturbance response.

3. Theory

Process Control is a subset of Control Theory, a branch of engineering and mathematics that manages the control of dynamic systems. Specifically, Process Control uses feedback loops to monitor, manage, and optimize processes. In industry, Process Control is used for manufacturing, chemical production, and refinement, ensuring optimal production, efficiency, and safety.

With any process, the *controlled variable*, *manipulated variable*, and *disturbances* are fundamental in understanding Process Control. The figure below shows the variables in tandem.

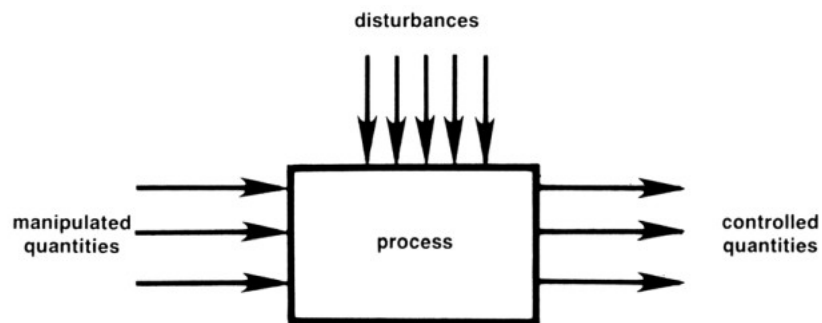


Figure 1. Process Control Variables

The controlled variables are the states or streams that will be controlled at a given level. This includes flow rates, levels, pressures, temperatures, or compositions. [3] For each of these variables, a *Set Point* is set for the controlled value. In the Process Control Lab, our Set Point is set to set to 75mm (default value). For each controlled variable,

there is a given manipulated variable and in most cases is the flow stream, with its respective flow rate managed with a control valve. With the Armfield PCT50, two valves are used. One is manually changed and the other is controlled by the software, with each valve being varied with different sized orifices. The solenoid valve is used to control the outflow with the opening and closing of the valve on a time cycle. [4]

Disturbances are unintended changes to the system that drive away the controlled variables from their set point. [3] To mitigate these changes, the system will automatically adjust the manipulated variables to keep the set point constant. In the case of the set point being changed, the manipulated variables will need to change to adjust to the controlled variables' new value.

A feedback control is used to automate the control of a process. Using sensors and other measuring devices, the values of the controlled variables can be measured. With the Process Control: Level Control lab, the Armfield PCT50 is used to understand water level control processes showing a clear understanding of a single PID (Proportional–integral–derivative) loop. The PCT50 has a level sensor with a range of 0 -150mm H₂O with the sensor mounted through the bottom of the acrylic tank. [4]

Using the PCT50, three different scenarios can be shown. In Experiment 1, the PCT50 will be controlled manually with an open loop. The pump speed will be adjusted to balance the water outflow and the solenoid adding disturbance. Experiment 2 is a closed loop demonstrates how changing the Proportional Band affects the speed and stability of the flow. Finally with Experiment 3, we will determine the effect of the different Integral Time settings at a constant P setting with the response and stability controlled using the P + I controller which varies the pump speed. With these different

scenarios, the water level, pump speed, Set Point, and the opening and closing of the solenoid valve, the effect of making changes to the overall system and controller settings will be shown with our applied disturbances showing how applying the correct system and controller settings will add stability.

4. Results

Experiment 1

This part of the lab involved manually controlling the pump speed manually to allow the water level to reach the designated set point. Several tests were performed, including filling, draining, and maintaining the water level in the tank with and without introducing a disturbance. The water level over the duration of these tests is shown below. At the initial speed, the flowrate into the tank was greater than that out of the tank, leading to a rise in water level at this speed setting.

Solenoid On

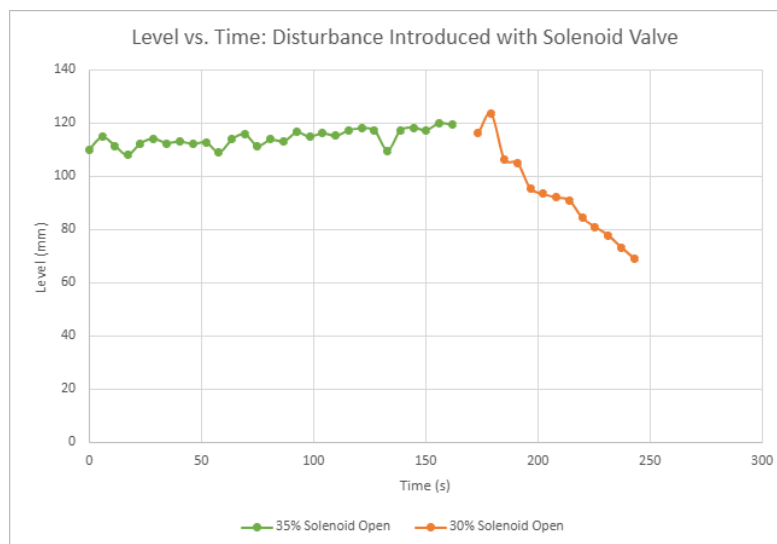


Figure 2. Water level with disturbance and open solenoid valve

Opening this valve increased the flowrate out of the tank, so increasing the pump speed was needed to maintain a steady level.

Solenoid Off

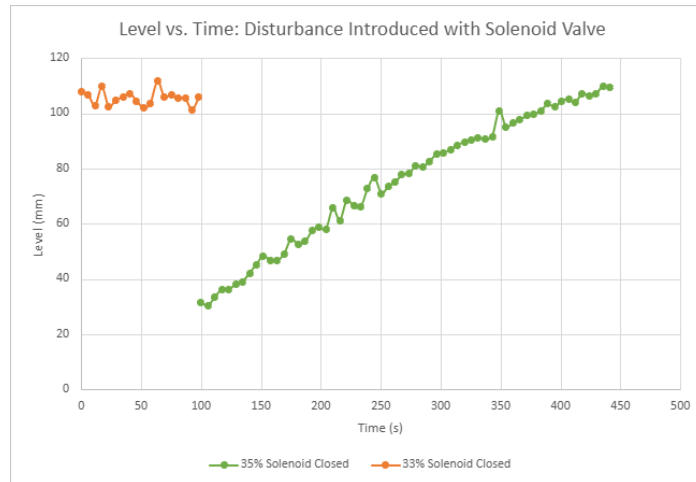


Figure 3. Water level with disturbance and closed solenoid valve

Closing the solenoid decreased the flow out of the tank, leading to a decrease in the pump speed needed to maintain a steady water level. Additionally, changes in the pump speed had a more immediate effect when the solenoid valve was open as opposed to being closed.

Filling at Initial (50%) and 75% Speed

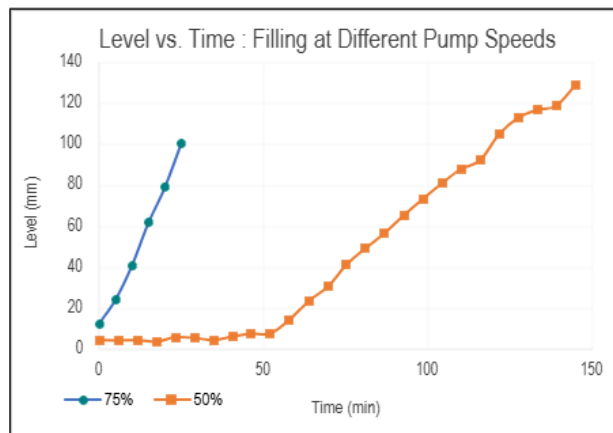
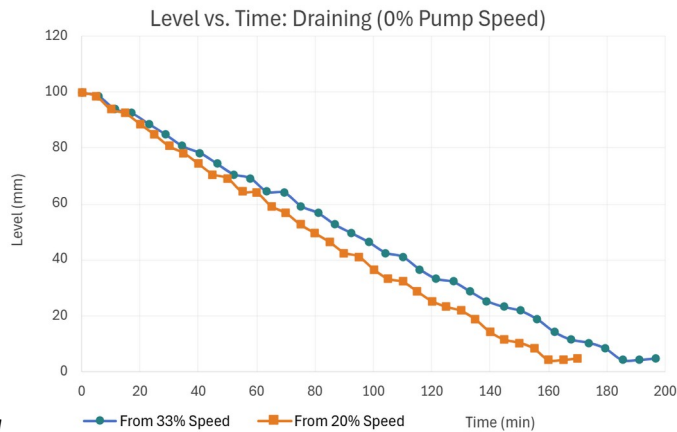


Figure 4. Water Level While Filling at 75% and 50% Pump Speed

Filling at both speeds was a similar process, although the level rose more quickly with a higher pump speed.



Draining

Figure 5. Water level while draining from initial speeds of 33% and 20%

As shown in **Figure 5. Water level while draining from initial speeds of 33% and 20%**, the water level during the draining process remained largely unchanged despite differences in starting pump speed. This is to be expected, as the outlet valve sizes were kept constant, and the pump speed was set to 0 in all instances for draining to occur. These tests establish a basis for the behavior observed when other controls are applied in later experiments.

Experiment 2

In experiment 2, the manual control used in the prior experiment is replaced by variable proportional band (P) control. The % represents the possible offset with respect

to the process variable (water level). Three tests were performed with only P-controls.

The first was completed with a setpoint water level of 75mm:

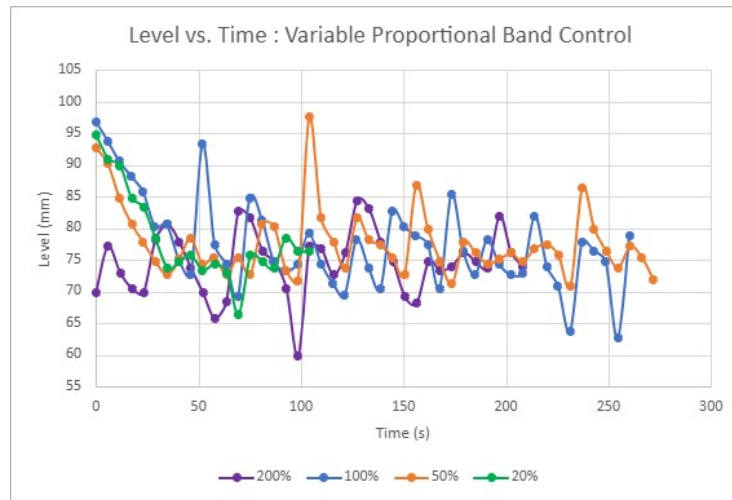


Figure 6. Water level using 200%, 100%, 50%, & 20% P settings with a 75mm setpoint

As shown in the figure above, at lower P settings, convergence to the desired setpoint occurred more quickly. However, there was still some oscillation above and below the setpoint at each tested value. This oscillation dampened over time at higher settings but remained more consistent at 50% and 20%. Despite this, the lower settings remained closer to the desired water level. These settings were further explored by introducing a disturbance from opening the solenoid valve. This case is shown below in *Figure 7*.

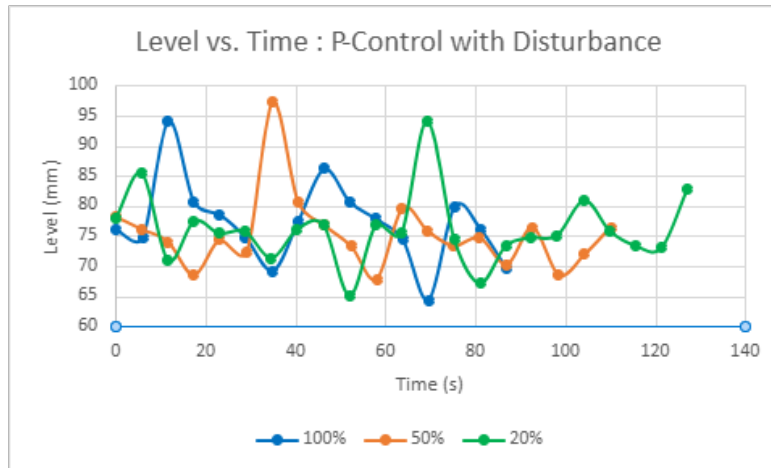


Figure 7. Water level using 100%, 50%, & 20% P controls with an open solenoid valve

Here in Figure 8, a similar trend is observed; however, the 20% value is less consistent than the 50% case and tends towards a water level that is above the setpoint. Notably, the 100% setting also reaches a water level close to the setpoint more quickly than the 50% setting, but with a greater offset. Finally, the original test with the solenoid valve closed was repeated at a higher setpoint (100mm).

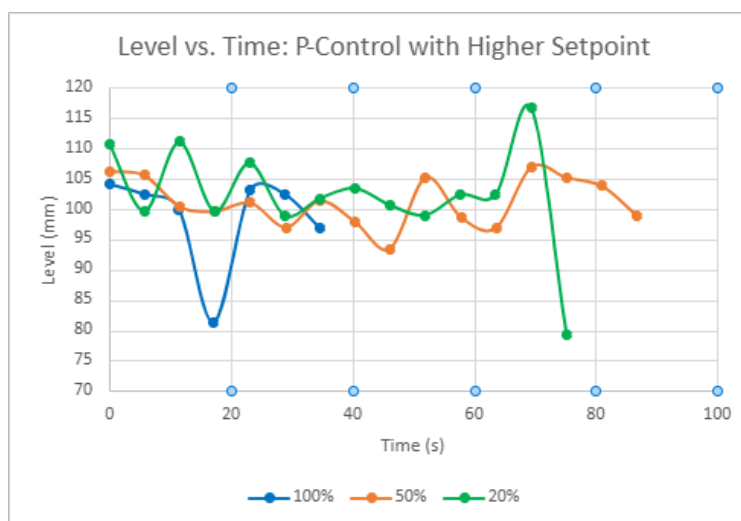


Figure 8. Water level using 200%, 100%, 50%, & 20% P settings with a 100mm setpoint

This is consistent with trends observed in the case with the introduced disturbance, although less data was collected for some trials in this experiment. All settings required less time to reach a value close to the setpoint than in trials with a lower desired water level. These tests verify the utility of proportional band controls and demonstrate the optimal value for maximizing stability and minimizing offsets with this piece of equipment. The higher 200% and 100% settings produce large offsets, while the lowest setting was unstable in some instances. Operating at the 50% setting seems to produce the best results, which can be verified with the tabulated offset values in Appendix I.

Experiment 3

In experiment 3, a constant proportional band setting of 20% was used with variable levels of integral control. This setting was chosen during experiment 2 due to its overall stability and relatively limited offsets. Since both 50% and 20% were viable options, the 20% option was chosen to see whether the somewhat higher instability could be mitigated with integral control. With this constant P-control, several different integral control (I) values were used to achieve the 75mm setpoint. These results can be seen below *Figure 9*.

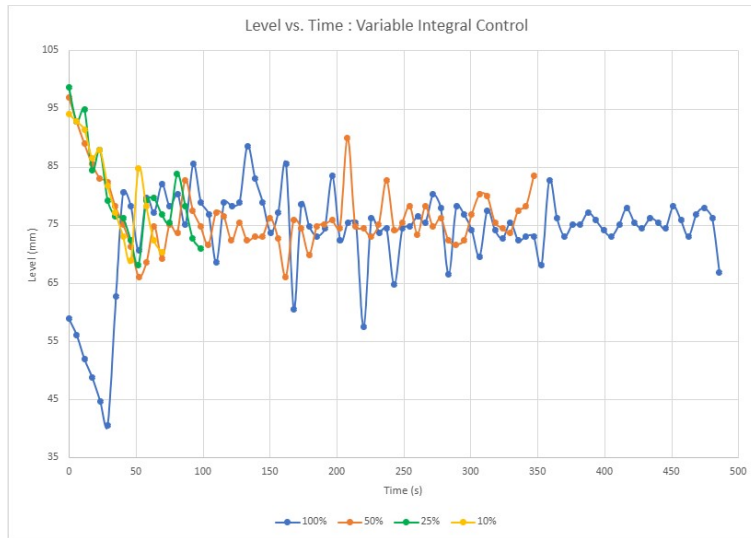


Figure 9. *Level vs. Time at Variable Integral Control*

At the 75mm set point, most of the I settings start with very similar offsets and stability. Although the data for some cases are somewhat limited, the graph *Figure 10* shows that higher settings converge to a value near the set point more quickly but are less stable once the set point is approximated. This is particularly clear in comparing the 100% and 50% cases. This can be further investigated, much like the P-only control in experiment 2. Introducing a disturbance produces the following results:

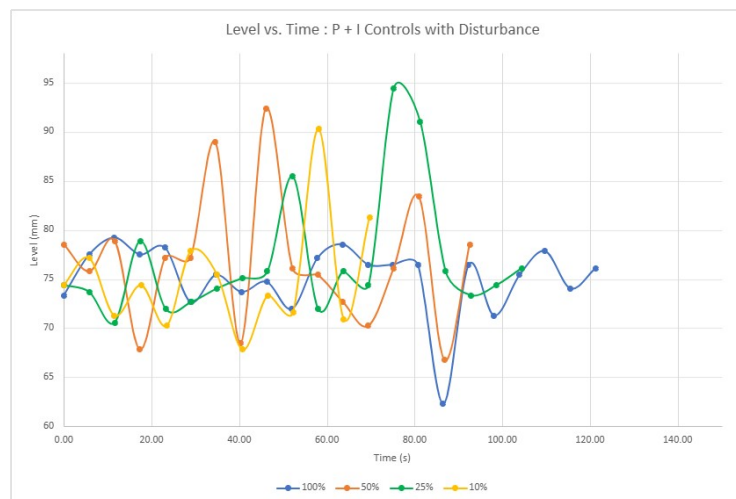


Figure 10. *Level vs. Time at PI Controls with Disturbance*

This test verifies the conclusions of the initial test, with the highest integral control being the least sensitive to disturbance. Finally, a test at a higher (100mm) setpoint was conducted:

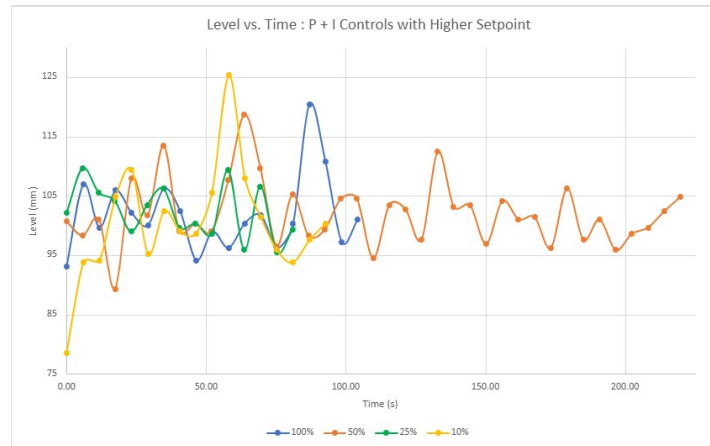


Figure 11. *Level vs. Time PI Controls with Higher Setpoint*

In addition to validating the trends observed in the previous tests, the higher setpoint demonstrates one of the drawbacks of a higher I settings, as in the 50% case the value begins to settle near a value that is greater than the actual set point. Interestingly, this is not observed at the 100% setting. Observing the offset in each instance [Appendix I] provides a better understanding of when the water level destabilizes. Notable destabilization is apparent in the 10% case.

5. Discussion

The results obtained from the level control experiments using the Armfield PCT50 system are consistent with fundamental process control theory for liquid level systems. The observed trends in response time, steady state offset, and oscillatory behavior across manual, proportional, and proportional-integral control models align with expected controller behavior describes in control theory literature [3]. In particular, changes in proportional band and integral action produced predictable effects on stability,

offset, and disturbance rejection, validating both experimental set-ups and underlying theoretical models [1,2].

During experiment 1 in manual control, changes in pump speed directly affected the inlet flow rate, resulting in predictable changes in the tank water level. When the inlet flow rate exceeded the outlet flow rate, the water level increased, whereas reducing the pump speed or opening the solenoid valve caused the level to decrease. This is because the behavior is consistent with a simple unsteady state mass balance on the tank, where accumulation is governed by the difference between inlet and outlet flow rates [3].

The draining experiments showed very minimal dependence on the initial pump speed, as the pump was turned off and the outlet valve configuration, like orifice size, remained constant. It is evident that, like theoretical expectations, the rate of level decreases is independent of the initial pump speed and more dependent on gravity-driven drainage [3].

In experiment 2, the proportional only control experiments demonstrated the expected tradeoff between speed, stability and steady state offset. As the proportional band (P) decreased from 200% to 20%, the system responded more rapidly to deviations from the setpoint, resulting in reduced offset. However, lower proportional band (P) settings also produced increased oscillations, particularly seen at the 50 and 20% setting, indicating reduced system stability. This behavior is consistent with classical proportional control theory, where higher controller gain improves responsiveness but can lead to oscillatory or unstable behavior [3].

When a disturbance was introduced by opening the solenoid valve, the proportional controller responded by adjusting the pump speed to counteract the

increased outlet flow rate. Although the system was able to reestablish a new steady state, a persistent offset remained, which is characteristic of p-only control and arises because proportional action alone cannot fully eliminate steady state error.

Experiment 3 introduces the addition of integral control (PI) which significantly reduced steady state offset compared to proportional only control, confirming the role of integral action in eliminating persistent error. Higher integral settings resulted in faster convergence toward the setpoint but came at the expense of increased oscillation and reduced stability. This tradeoff is well documented [1,2] in control theory, as excessive integral action can lead to overshoot and sustained oscillations.

At the higher set point of 100 mm, some integral settings caused the system to settle above the desired level, suggesting overcompensation by the integral term. This behavior highlights the importance of careful controller tuning and demonstrates that controller parameters optimized for one operating condition may not be optimal across all setpoints.

Several limitations in the experimental setup may have influenced the results. Sensor noise and resolution limits in the level measurement likely contributed to smaller oscillations near the setpoint. Additionally, delays in valve response and pump speed adjustments may have been introduced lag into the system response. The pump should also be inspected prior to the start of the experiments to ensure it is working properly. Future experiments may benefit from finer controller tuning increments and improved calibration of the level sensor could also reduce measurement uncertainty.

From these experiments it is evident that proportional integral (PI) control provides superior offset elimination compared to proportional (P) only control for liquid

level regulation. Conclusions for the optimal controller settings are not as definitive, as the tuning parameters were specific to this experimental system and operating conditions. These results reinforce the importance of controller tuning in industrial level control applications, where improper tuning can lead to instability. Future work should explore controller performance under varying inlet flow disturbances to better replicate industrial conditions.

6. References

1. Fellani, M. A., & Gabaj, A. M. "PID controller design for two tanks liquid level control system." *International Journal of Electrical and Computer Engineering* (2015): 436~442 . <https://ijece.iaescore.com/index.php/IJECE/article/view/5637>.
2. 2. Urrea, Claudio and Felipe Paez. "Design and Comparison of Strategies for Level Control in a Nonlinear Tank System." *Processes* (2021): 9(5), 735. <https://doi.org/10.3390/pr9050735>.
3. Murrill P. W., "Fundamentals of Process Control Theory," Unit 2, 3rd Edition, International Society of Automation (ISA), Research Triangle Park, NC, 2000: https://www.isa.org/getmedia/62691c17-de64-4854-8eb4-f413d1ae2fa4/fundamentals-of-process-control-theory-thirded_murrill_unit2.pdf
4. Armfield Ltd., "PCT50 Process Control Trainer – Datasheet," Version v3b, Armfield Limited, Ringwood, Hampshire, UK, 2019: https://armfield.co.uk/wp-content/uploads/2019/12/Armfield-PCT50-Datasheet-v3b_Web.pdf
5. Murrill P. W., "Fundamentals of Process Control Theory," Unit 2, 3rd Edition, International Society of Automation (ISA), Research Triangle Park, NC, 2000:

[https://www.isa.org/getmedia/62691c17-de64-4854-8eb4-f413d1ae2fa4/
fundamentals-of-process-control-theory-thirded_murrill_unit2.pdf](https://www.isa.org/getmedia/62691c17-de64-4854-8eb4-f413d1ae2fa4/fundamentals-of-process-control-theory-thirded_murrill_unit2.pdf)

6. ChE 382: Unit Operations Laboratory. (2026). Process Control: Level Control (*Rev. Spring 2025v1*). University of Illinois Chicago, Department of Chemical Engineering.

7. Appendix I: Data Tabulation/Graphs

8. Appendix II: Error Analysis

This experiment was performed on the Armfield PCT50 level-control apparatus, where water level was measured using an electronic pressure sensor and manipulated through pump speed. Disturbances are introduced via a solenoid-controlled outlet. The primary measured outcomes were steady-state level, offset, and fill/drain flow behavior derived from level vs. time data. These quantities are directly affected by measurement noise, hardware configuration, and controller behavior.

Both random (precision) and systematic (bias) uncertainties influence whether trends in offset and stability can be confidently attributed to controller tuning rather than experimental instrumentation.

For each controller setting, the system was allowed to reach steady conditions before data was analyzed. A segment of time where the level appeared constant was selected, and the average level in this region was used to represent the steady state value. However, even when the level appears visually stable, small fluctuations were always present due to sensor noise, minor turbulence in the tank, and pump pulsation.

The spread of the level readings in this steady region provides a measure of measurement variability. From this variability, a standard error and confidence limit can be estimated to show the range within which the true average level likely falls.

Because offset is calculated directly from the measured level, the same uncertainty applies to the offset value. Reporting the steady-state level and offset together with confidence limits demonstrates how certain it is that P-only control produces a real,

measurable offset and that PI control reduces or eliminates that offset. Without including this uncertainty, it would be unclear whether small differences in level were caused by controller changes or simply by measurement noise.

Furthermore, flow rate is not measured directly, but inferred from how quickly the level in the tank rose or fell during the filling and draining. This contains uncertainty related to small fluctuations in level readings, human errors in reading the steady state region, and measurement uncertainty in tank dimensions used to relate level change.

Overall, there are three major sources of experimental error. The level sensor required zeroing when the tank is emptying. If this was not done correctly, all levels would be shifted by a constant amount. This directly affects the calculated offset and could make the controller appear to perform better or worse than it did. Air in the pipeline could also cause experimental error, as if the pump line contained air, the delivered flow rate would be lower than expected for a given pump speed. This changes the dynamics of the system and makes the comparison of the controller settings less reliable. Judgment then the process reached steady state could also produce error, as visual determination was used to see if the system reached steady state, which could cause error in the level reading the offset.

Including quantitative uncertainty alongside qualitative error analysis strengthens confidence in the experimental conclusions. While measurement noise and setup variability influenced the precision of the results, they did not obscure the fundamental control principles demonstrated in the experiment. Improved standardization and repeated trials would further reduce uncertainty and improve reliability of the data.

9. Appendix III: Sample Calculations

Calculation of offset for experiment :

10. Appendix IV: Individual Team Contributions

<i>Table 1: Individual Team Contributions</i>	
Contribution	Team Member
Executive Summary, Introduction	Angelica Ramirez
Theory	Noah Guale
Results	Basil Hashim
Discussion	Jazmin Aca
Appendix I, II	Hiba Naqvi
References	All